Problem 65 (JBMO Shortlist 2017 C3)**.** We have two piles with 2000 and 2017 coins respectively. Ann and Bob take alternate turns making the following moves: The player whose turn is to move picks a pile with at least two coins, removes from that pile t coins for some $2 \leqslant t \leqslant 4$, and adds to the other pile 1 coin. The players can choose a different t at each turn, and the player who cannot make a move loses. If Ann plays first determine which player has a winning strategy.

Solution. Bob has a winning strategy. We claim that at any moment, if the two piles contain a and b coins respectively, the next player has a winning strategy if and only if $a - b \equiv 2, 3, 4, 5, 6 \pmod 8$.

One cannot move when $(a, b) = (1, 1), (1, 0), (0, 1), (0, 0)$. The claim holds in these base cases. Suppose the claim holds for all (a, b) with $a + b \leqslant k - 1$. Consider the case (a, b) with $a + b = k$ different from the base cases. WLOG assume $a \geqslant b$.

- If $a - b \equiv 2, 3 \pmod 8$, then $a \geqslant 2$. The next player P can remove 2 coins from the first pile, so that the new difference becomes $(a - 2) - (b + 1) \equiv -1, 0 \pmod 8$. Since the number of coins is decreased, and P is now the second player, P has a winning strategy by the claim.
- If $a - b \equiv 4, 5, 6 \pmod 8$, then $a \geqslant 4$. The next player P can remove 4 coins from the first pile, so that the new difference becomes $(a - 4) - (b + 1) \equiv -1, 0, 1 \pmod 8$. By the claim, P has a winning strategy.
- If $a - b \equiv 0, 1, 7 \pmod 8$ and the next player removes $2, 3$ or 4 coins from the second pile (the pile with the smallest number of coins), the other player Q can remove $2, 3$ or 4 coins from the first pile. The difference is unchanged, and Q has a winning strategy by the claim.
- If $a - b \equiv 0, 1, 7 \pmod 8$ and the next player removes $2, 3$ or 4 coins from the first pile (the pile with more number of coins), the other player Q can remove $4, 3$ or 2 coins from the first pile unless $a \leqslant 5$. If this can be done, the new difference is $(a - 6) - (b + 2) \equiv 0, 1, 7 \pmod 8$. By the claim, Q has a winning strategy. It remains to consider the following cases.

$$\begin{aligned}
&(5,4) \longrightarrow (5-r,5) \longrightarrow (6-r,5-r)\\
&(4,3) \longrightarrow (4-r,4) \longrightarrow (5-r,4-r)\\
&(3,2) \longrightarrow (1,3),(0,3) \longrightarrow (2,1),(1,0)\\
&(2,1) \longrightarrow (0,2) \longrightarrow (1,0)
\end{aligned}$$

  In all cases, the difference is unchanged, and Q has a winning strategy by the claim.

Now, since $2017 - 2000 = 17 \equiv 1 \pmod 8$, the second player Bob has a winning strategy. $\square$

Problem 66 (Rioplatense MO 2013 Level 3 Problem 4)**.** Two players A and B play alternatively in a convex polygon with $n \geqslant 5$ sides. In each turn, the corresponding player has to draw a diagonal that does not cut inside the polygon previously drawn diagonals. A player loses if after his turn, one quadrilateral is formed such that its two diagonals are not drawn. A starts the game.

For each positive integer n, find a winning strategy for one of the players.

Solution. We claim that A has a winning strategy if and only if n is even.

Since the diagonals cannot cross each other, drawing a diagonal is the same as cutting the polygon into two pieces. One cannot move when all remaining polygons are triangles (with no diagonals) and pentagons (a quadrilateral is formed no matter how one cuts it). Note that in each move, one more polygon is formed and the parity of the total number of sides is unchanged (the same as n). If there are k triangles and ℓ pentagons at the end, then we must have $k + \ell \equiv 3k + 5\ell \equiv n \pmod 2$. This shows the total number of polygons at this stage has a fixed parity depending on the parity of n. If n is even, it must be B's turn. Otherwise, it must be A's turn. This proves the result. And the player who has a winning strategy can play in whatever way as long as quadrilaterals are not drawn. $\square$

Problem 67 (Iran TST 2018 Test 2 Problem 2)**.** Mojtaba and Hooman are playing a game. Initially, Mojtaba draws 2018 vectors with zero sum. Then in each turn, starting with Mojtaba, the player takes a vector and puts it on the plane. After the first move, the players must put their vector next to the previous vector (the beginning of the vector must lie on the end of the previous vector). At last, there will be a closed polygon. If this polygon is not self-intersecting, Mojtaba wins. Otherwise Hooman wins. Who has the winning strategy?

Solution. Mojtaba has a winning strategy. Consider $\triangle ABC$. Suppose Mojtaba draws the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$ and 2016 copies of $\mathbf{v} = \dfrac{1}{2016}\overrightarrow{CA}$. Firstly, Majtaba chooses $\mathbf{v}$. Every time when Hooman chooses $\mathbf{v}$, Majtaba chooses $\mathbf{v}$ again unless all $\mathbf{v}$'s are used, in which case Majtaba can choose any of the remaining two vectors. When Hooman chooses $\overrightarrow{AB}$ or $\overrightarrow{BC}$, Majtaba chooses the other. At the end, since $\overrightarrow{AB}$ and $\overrightarrow{BC}$ must be placed next to each other, the resultant polygon must be a triangle congruent to $\triangle ABC$. Thus, it is not self-intersecting, and Mojtaba wins. $\square$

Problem 68 (Austrian MO 2017 Problem 3)**.** Anna and Berta play a game in which they take turns in removing marbles from a table. Anna takes the first turn. When at the beginning of the turn there are $n \geqslant 1$ marbles on the table, then the player whose turn it is removes k marbles, where $k \geqslant 1$ either is an even number with $k \leqslant \dfrac{n}{2}$ or an odd number with $\dfrac{n}{2} \leqslant k \leqslant n$. A player wins the game if she removes the last marble from the table.

Determine the smallest number $N \geqslant 100000$ such that Berta can enforce a victory if there are exactly N marbles on the tale in the beginning.

Solution. The answer is $2^{17} - 2 = 131070$. We claim that the second player has a winning strategy if and only if the number of marbles remaining is of the form $2^m - 2$ for some integer $m \geqslant 2$. We prove this by induction.

If there is only 1 marble left, the first player must win. If there are 2 marbles left, the first player can only remove 1 marble, so the second player wins.

Suppose there are a marbles left. If a is odd, the first player wins by removing all marbles. If a is even and there exists $m \geqslant 2$ such that $2^m \leqslant a \leqslant 2^{m+1} - 4$, the first player can remove $a - (2^m - 2)$ marbles since this number is even and $a - (2^m - 2) \leqslant \dfrac{a}{2}$. The number of marbles remaining is $2^m - 2$, so the first player wins by the inductive hypothesis. If $a = 2^m - 2$ for some integer $m \geqslant 2$, then the number of marbles remaining after the next move can only be

$$2^m - 4, 2^m - 6, \ldots, 2^m - (2^{m-1} - 2), 2^{m-1} - 1, 2^{m-1} - 3, \ldots, 1.$$

All these numbers are not of the form $2^\ell - 2$, and so the second player wins by the inductive hypothesis. So the claim is proved.

Since $2^{16} - 2 = 65534 < 100000$ and $2^{17} - 2 = 131070 > 100000$, the smallest N is 131070. $\square$